

## AERO-TWIN

From Reactive Alarms to Predictive Action to AI That Saves UAVs 30 Minutes Before Failure

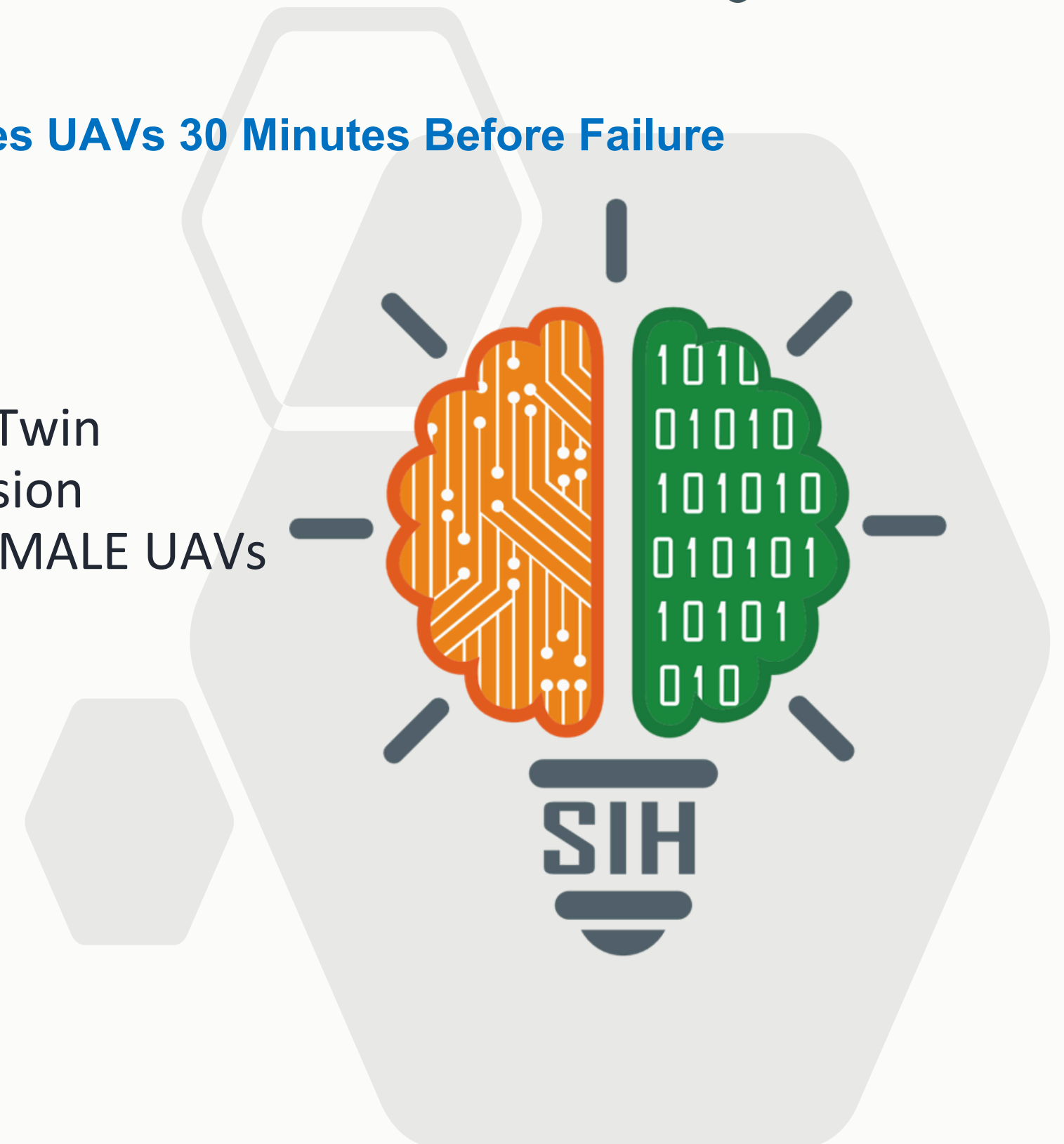
**Problem Statement ID:** 26054

**Problem Statement Title:** AI-Enabled Real-Time Digital Twin System for Health Monitoring, Fault Prediction and Mission Reliability Enhancement of Aero Piston Engines used in MALE UAVs

**Theme:** Robotics and Drones

**PS Category:** Software

**Team Name:** Midnight Ciphers



### THE PROBLEM

- Current monitoring is **threshold-based** and **reactive**, detecting faults only after abnormalities occur.
- MALE UAVs (TAPAS-BH-201, Rustom-II ) fly 18-30+ hour missions : **engine loss = total aircraft loss**
- **No capability to estimate Remaining Useful Life (RUL)** or predict degradation trends
- **No synchronized** digital mirror to simulate engine behavior under varied conditions
- **Ladakh's thin air** and **Thar's extreme heat** create **distinct engine stress** and **failure patterns**.

### OUR SOLUTION

- **Indigenous, modular** Digital Twin with a continuously synchronized virtual replica of the **Rotax 912 iS** engine.
- Real-time **anomaly detection, RUL prediction, mission replay & explainable fault diagnosis**.
- Moves from threshold-reactive to predictive diagnostics - detects faults **30-60 min early**
- **20 Hz** physics twin predicts engine behavior, residuals reveal degradation.
- 100% Air-Gapped - **Offline** AI & analytics with **zero cloud dependency**.

### HOW IT ADDRESSES THE PROBLEM

- **AI catches degradation early** : Enables **proactive** maintenance before failure occurs
- **Digital Twin mirrors physical engine** : Physics-based model **synchronized with live data**
- **Mission Simulation**: Simulates **Ladakh** (22,000 ft) & **Thar Desert** (+48°C) - validates operations before deployment
- **Complete mission lifecycle insight** : Monitors **real-time health** during flight
- **Actionable diagnosis** : Diagnostic time reduced from **hours to minutes**

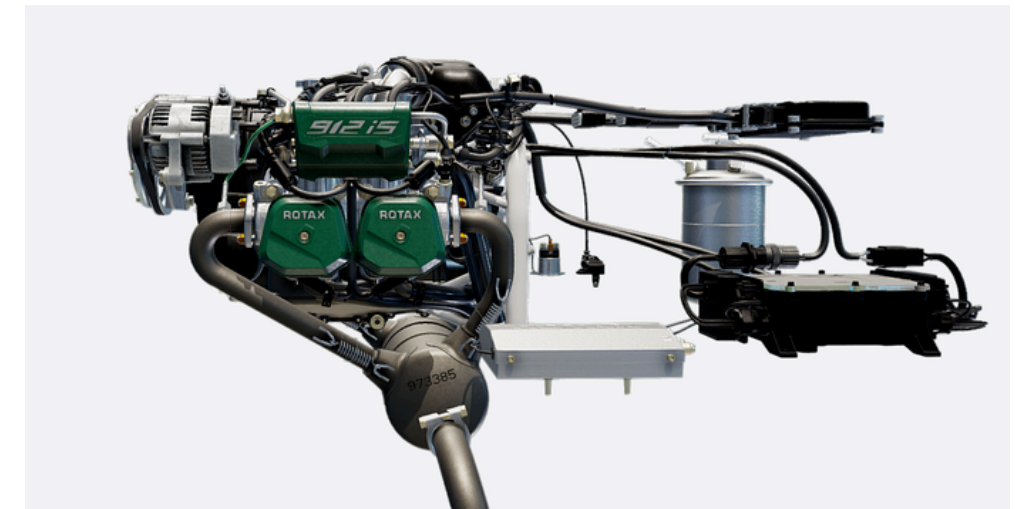
### INNOVATION & UNIQUENESS

- **Engine-Agnostic**: One framework, any **engine - adaptable** to indigenous and imported engines
- **Indigenous**: Built from first principles - supporting **Aatmanirbhar Bharat**
- **Secure Telemetry**: **TLS + API Key** authentication extendable to mTLS for defence-grade security
- **Hybrid Physics + Data-Driven**: Physics provides interpretability; AI provides **adaptability**
- **Explainable AI**: every alert backed by a physical causal chain, not just a flag



**TAPAS-BH-201(Rustom-II)**

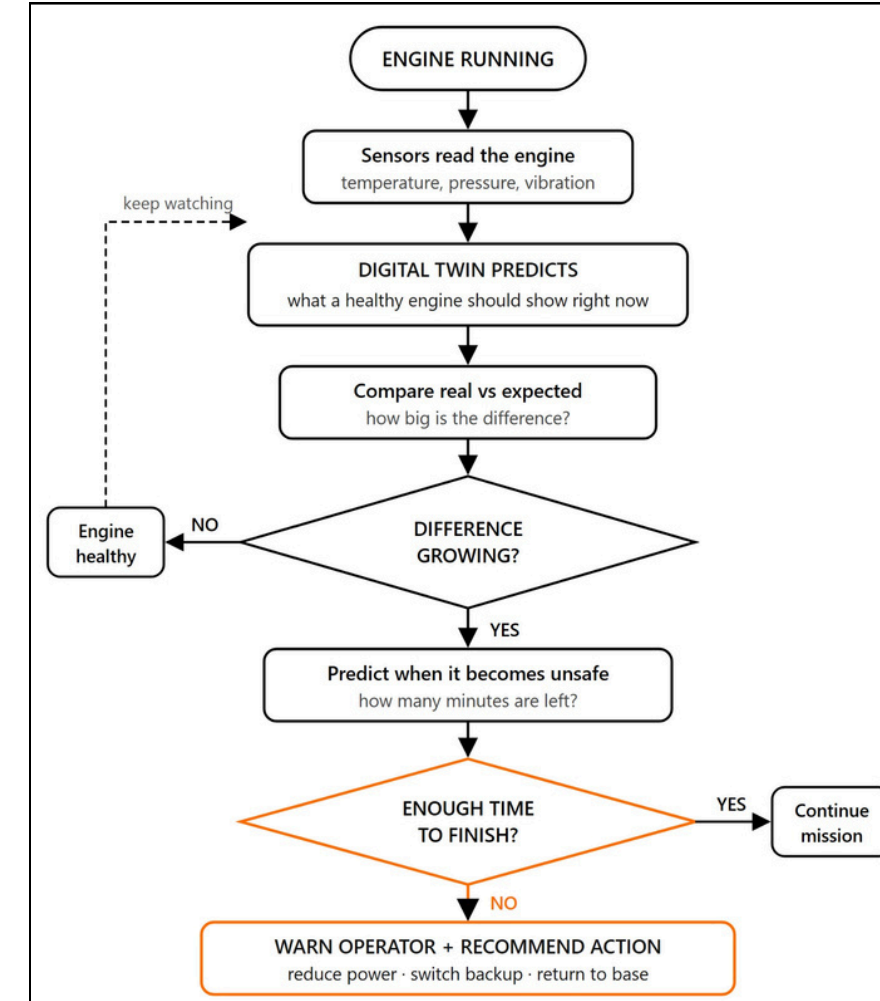
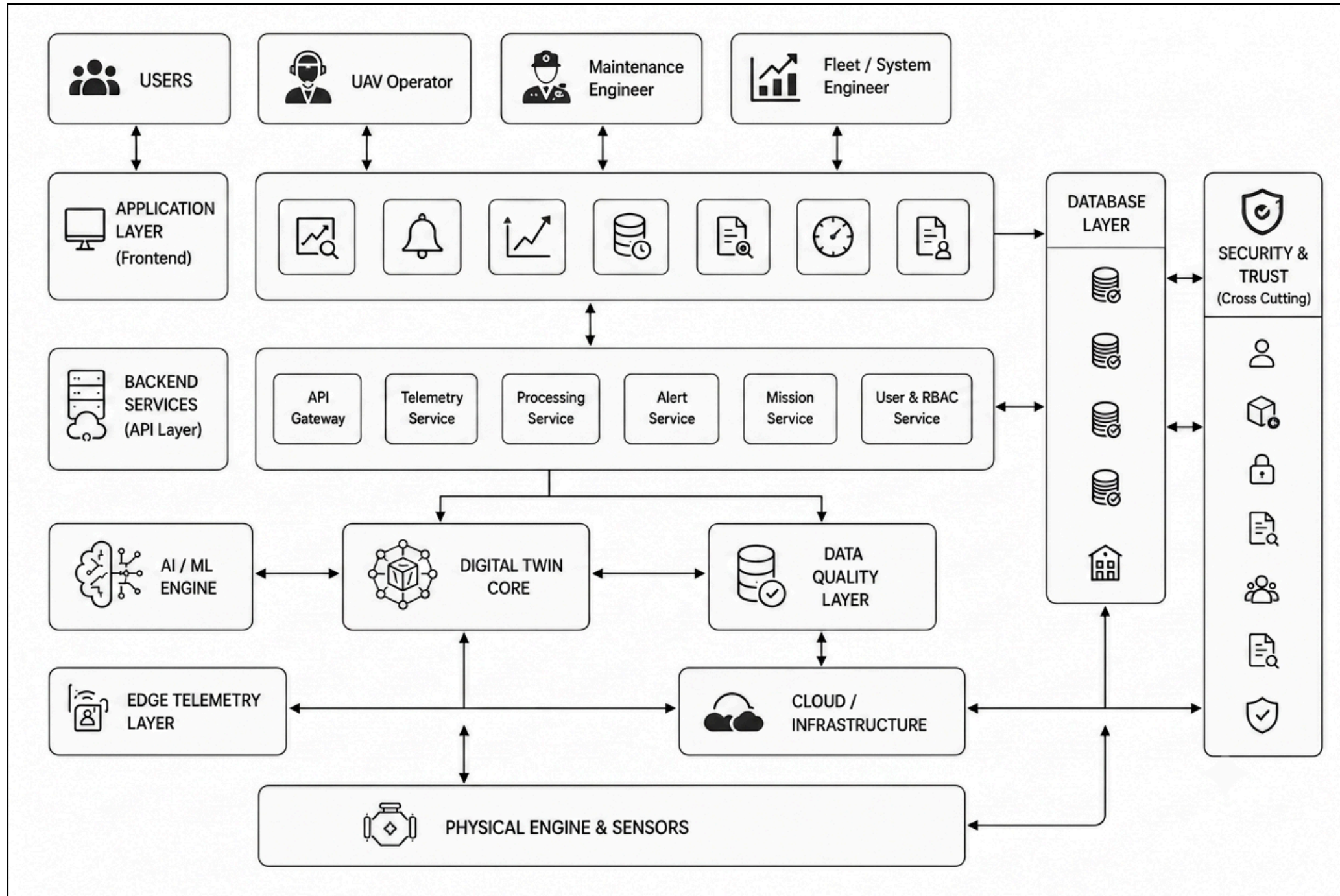
- **Indigenous MALE UAV** developed by DRDO.
- Endurance : **20 to 40+** hours
- Operational Radius : **150 – 300+** km
- Operating Altitude: **28,000–30,000** feet



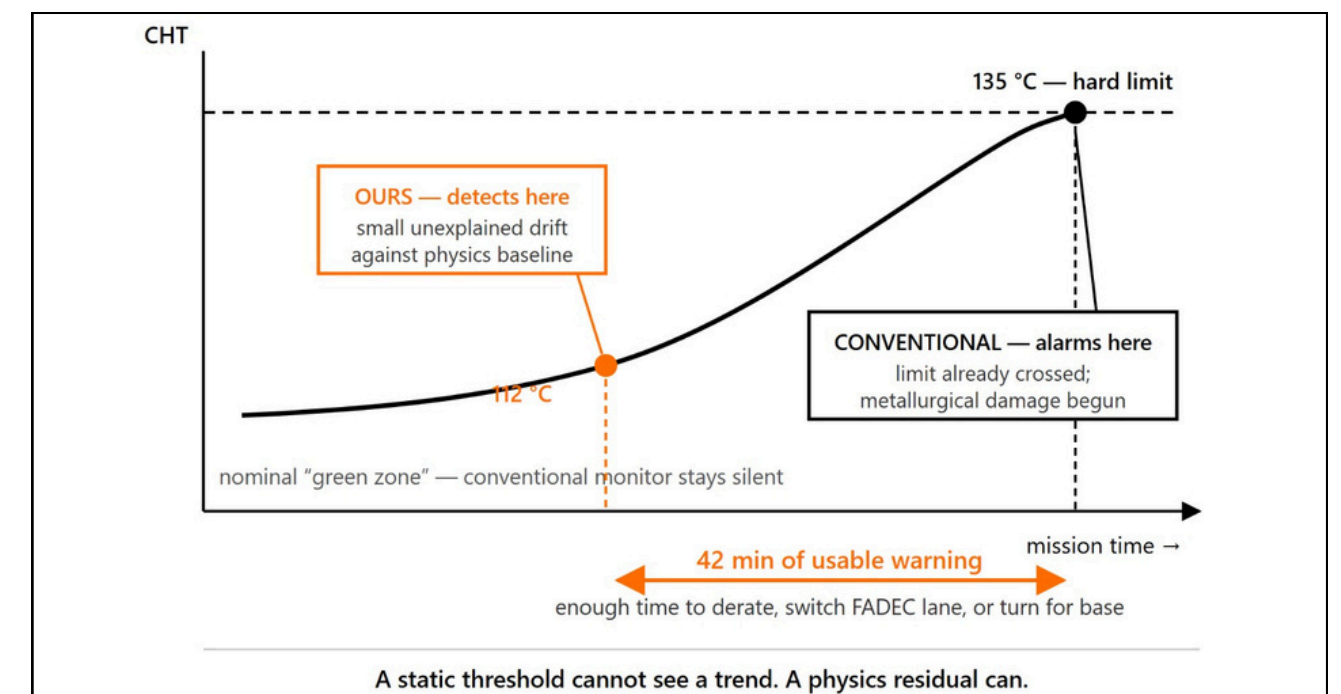
**Rotax 912 iS Sport**

- **4 cylinder** horizontally opposed aero piston engine
- Max RPM : **5,800**
- Takeoff Power: **100 HP**

## System Architecture



## TECH STACK



## FEASIBILITY

- **Technical:** End-to-end prototype running: **CAN ingestion, physics model, ML** and dashboard fully validated
- **Operational:** **AI recommends actions** while operators make final decisions, **reducing adoption friction**
- **Security:** All models, databases, and documents **stored locally**, ensuring **defence-grade** operational security
- **Economic:** Early RUL warnings prevent in-flight engine failures, **protecting high-value assets** from total loss

## VIABILITY

- **Market Need:** With over **500 MALE UAVs** globally and each needing propulsion health monitoring, the addressable **market is substantial**.
- **Engine-agnostic framework :** Framework **adapts to any aero-piston engine** by swapping CAN database, performance maps, and fault library
- **Technology Sovereignty:** **100% indigenous** IP with no foreign licensing requirements and no export restrictions
- **Scalable Architecture:** Modular design supports additional **fault modes, new engine types, and more UAV platforms**

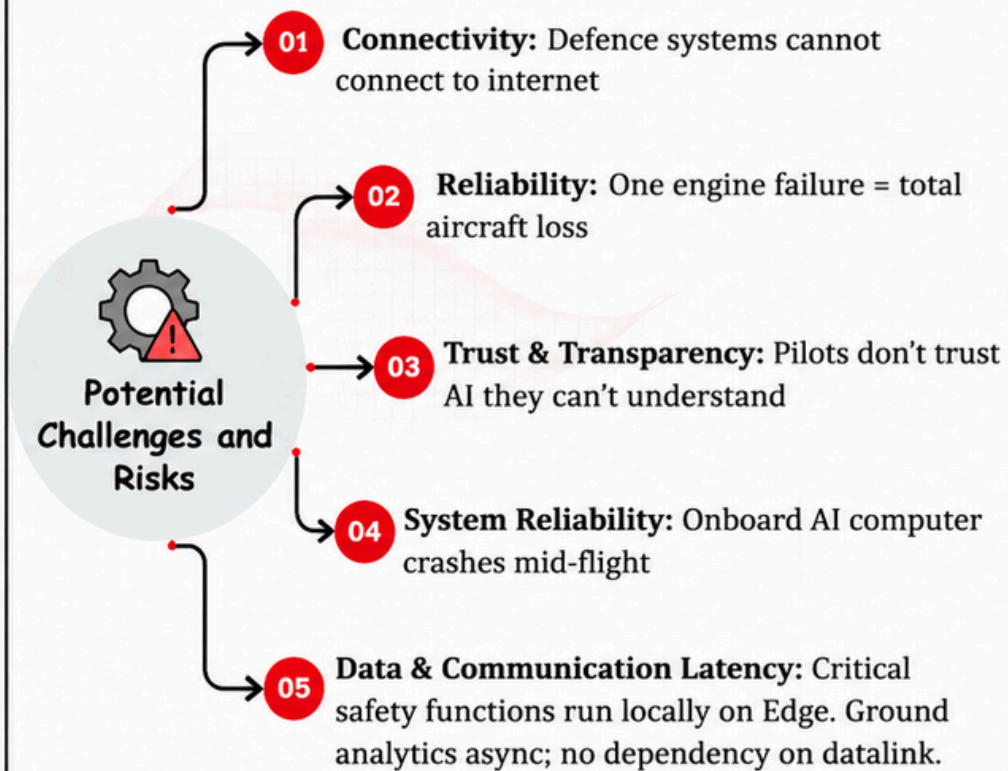
## INDUSTRY VALIDATION

Visited DRDO Pune for direct Interaction with DRDO Scientists  
Date: 1st September 2026

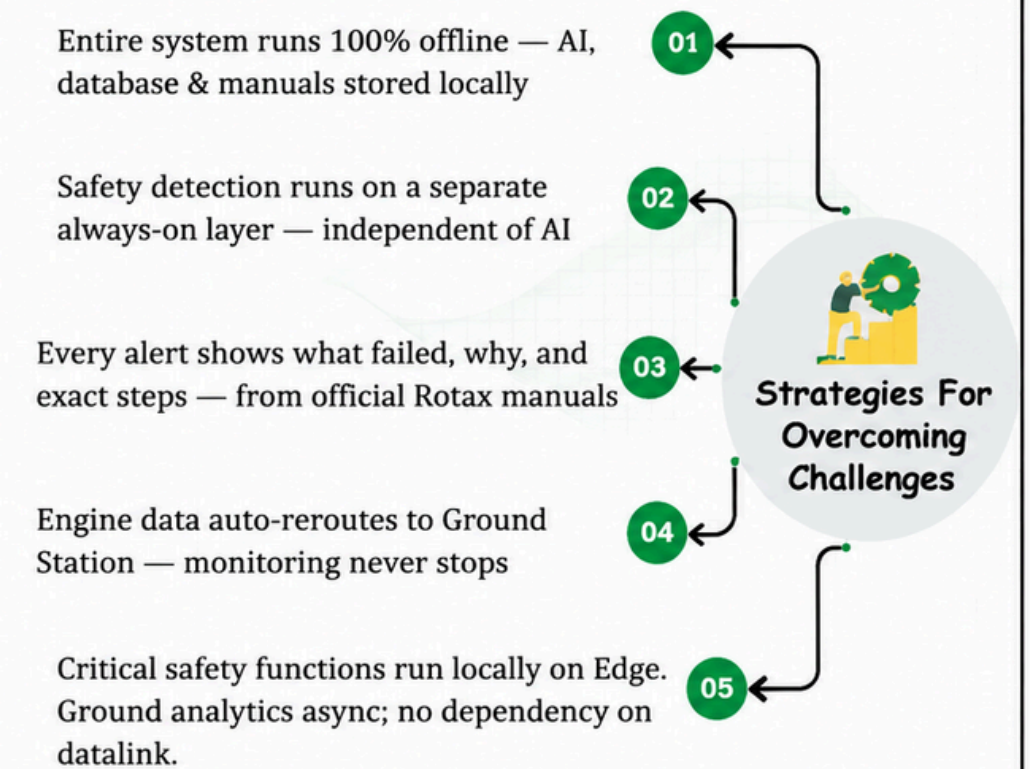
### KEY INSIGHTS VALIDATED ( Non-Confidential )

- Relevance of **DRDO fault scenarios** to actual operations
- Importance of **air-gapped deployment** for defence systems
- Criticality of early warning (**30-60 min horizon**)
- Need for **explainable AI** (pilot trust is paramount)
- Value of **mission replay** for **post-flight analysis**

### Potential Challenges and Risks



### Strategies to Overcome Challenges



## IMPACT

- **UAV Operators (Pilots):** Real-time health status and predictive alerts give operators **actionable intelligence**, enabling preventive **action before failures** occur
- **Defence Procurement:** Indigenous digital twin **reduces foreign OEM dependency** and directly supports Aatmanirbhar Bharat
- **Training & Safety:** Mission replay and post-flight analysis support **root cause investigation, enhanced training, and robust safety protocols**
- **Strategic Defence:** Border surveillance in **Ladakh** and **maritime patrol** in the Indian Ocean become possible with continuous **24/7 ISR** without propulsion failures
- **Global Defence Export:** **Scalable solution** for international MALE UAV fleets creates significant **potential export revenue**

## FUTURE POTENTIAL

1. **National Defence Network:** Integrates with **Indian Air Force, Indian Army, and Indian Navy** command and control systems for unified operational awareness
2. **Digital Forensics:** **Enables** post-crash analysis with **last-moment replay** and **Identifies failure causal** chains for future design improvements
3. **Autonomous Emergency Landing:** Digital Twin simulates remaining flight time during critical faults to **maximize asset recovery**
4. **Fleet-Wide Federated Learning:** Each UAV's local twin learns degradation patterns independently and Aggregated model weights create **fleet collective intelligence**



## BENEFITS OF THE SOLUTION



## ECONOMIC

**Reduced maintenance costs:**  
Predictive maintenance reduces unscheduled overhauls.



## STRATEGIC

**Indigenous capability:**  
Reduces dependency on foreign OEMs and aligns with India's defence indigenization goals.



## OPERATIONAL

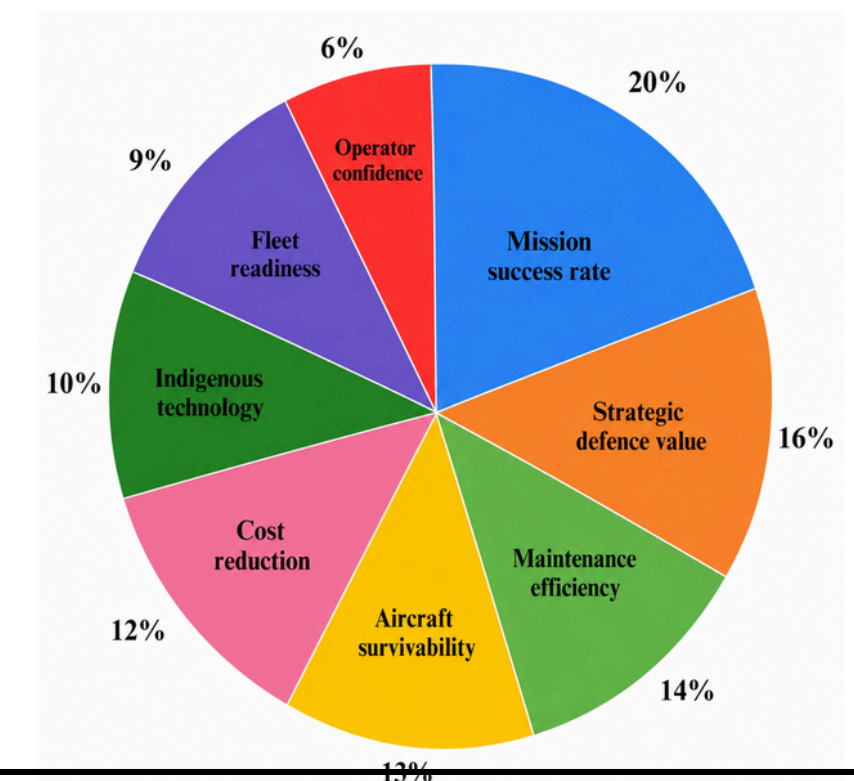
**Faster diagnostics:**  
Fault isolation matrix reduces diagnostic time from hours to seconds.



## ENVIRONMENTAL

**Extended engine life:**  
Less premature replacement = reduced manufacturing footprint.

## PROPORTIONAL IMPACT ACROSS DOMAINS



- **BRP-Rotax. Rotax 912 iS Sport Technical Documentation :** [Rotax 912 iS Sport](#)  
(Used for engine architecture, specifications, and performance data.)
- **MGL Avionics. RDAC-CAN 912iS Installation Manual. MGL :** [MGL Avionics Documentation](#)  
(Used for Rotax 912 iS ECU/CAN telemetry and engine-parameter acquisition.)
- **Kanardia. Daqu User & Installation Manual – Rotax iS CAN :** [Kanardia Daqu Manual](#)  
(Used for Rotax iS CAN-bus connectivity and telemetry validation.)
- **Linux Kernel Documentation. SocketCAN :** [Linux SocketCAN Documentation](#)  
(Used for CAN interface, virtual CAN and software-based telemetry acquisition.)
- **Heywood, J. B. Internal Combustion Engine Fundamentals, 2nd Ed. McGraw-Hill, 2018 :** [McGraw Hill – Internal Combustion Engine Fundamentals](#)  
(Used for engine-cycle, thermodynamic and thermal modelling foundations.)
- **EASA. Airworthiness Directives :** [EASA Safety Publications](#).  
(Used for aviation safety, maintenance and real-world engine failure considerations.)

[Research Documentation : click here](#)

| Research Area        | AeroTwin Approach                                       | Engineering Outcome            |
|----------------------|---------------------------------------------------------|--------------------------------|
| Engine Modelling     | Rotax 912 iS operating parameters + thermodynamic model | Expected engine behaviour      |
| Telemetry            | CAN / ECU data + SocketCAN + data-quality validation    | Reliable real-time telemetry   |
| Digital Twin         | Physics model + engine maps + virtual CHT               | Synchronized virtual engine    |
| Anomaly Detection    | Operating-regime-aware residual analysis                | Early & fewer nuisance alerts  |
| Fault Diagnosis      | Fault Isolation Matrix + parameter relationships        | Probable cause + evidence      |
| Mission Intelligence | Mission replay + what-if simulation                     | Health-aware mission decisions |
| Validation           | Fault injection + sensor failure scenarios              | Testable & measurable PoC      |